

An Introduction to Deeplearning4j, the Distributed Deep Learning Library

There are many deep learning libraries that are becoming popular among the developer community such as Theano, Torch, Caffe, etc. Deeplearning4J is an open source and distributed deep learning library targeted at Java Virtual Machine (JVM). This article provides an introduction to its capabilities and features.



Machine learning has led to a tremendous shift in the way we create algorithms to solve problems. It has made developers and researchers shift away from the traditional step-by-step solutions approach to a holistic model, which in many aspects mimics the methods that biological organisms employ while solving problems or acquiring a new skill. These days, machine learning models are employed in all those sectors that were originally difficult to deal with by the traditional algorithmic approaches. Real-time object detection from videos, classification and prediction systems are some of the tasks performed with the help of machine learning. The popular quote by Prof. Tom Mitchell, who is a pioneer in shaping the domain, clearly defines machine learning as follows: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E ."

Deep learning

Deep learning is a recent improvement in the machine learning domain. A basic requirement of machine learning is to identify and build a feature set. This task is generally carried out by experts in that domain, manually. So, for each problem domain, customised features need to be built by expert humans. This often creates a bottleneck in the overall process flow.

Deep learning differs from traditional machine learning in the way features are built. It attempts to build the features automatically from the given *large* data set. Note the emphasis on the word 'large'. For deep learning to work at a decent accuracy level, we need considerably large sized data sets. This is significant, because the machine requires a fairly large amount of data in order to detect the discriminative features of a set. Feature engineering is an important component for the successful implementation of any machine learning related project.